RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 16

Topic: Review the Use of Graphing Calculators

In this revision, we will review the *use of graphing calculators* for the following topics. There may be tricks that you may not be aware of.

1: Graphing – Functions
2: Graphing – Parametric Equations
3: Equations & Inequalities
4: Complex Numbers
5: Discrete Random Variables

1: Graphing – Functions

Skills Required: By using the **FUNCTION** mode,

1. Sketch the graphs of functions and composite functions [Y=, WINDOW, ZOOM].
2. Determine the y-value for a given x-value [[CALC] value].
3. Determine the x-intercepts [[CALC] zero].
4. Determine the coordinates of the minimum & maximum points [[CALC] minimum/maximum]
5. Determine the points of intersection of 2 graphs [[CALC] intersect].
6. Determine the gradient of the function [[CALC] $\frac{\mathrm{d}y}{\mathrm{d}x}$ OR nDeriv].
7. Determine the area bounded by the function with the x-axis [[CALC] $\int \mathrm{f}(x)\,\mathrm{d}x$ OR fnInt].
8. Sketch piecewise functions (OS 5.3.0 and 5.3.1) [MATH B: piecewise]

Exercise 1

(a) Functions f and g are defined as follows:

$$\mathrm{f}(x)=\frac{x^2-2x-1}{x+2}, \qquad \text{for } x\in\mathbb{R},\ x\neq -2,$$

$$\mathrm{g}(x)=x-3-\mathrm{e}^{-x}-\ln(x+1), \qquad \text{for } x>-1.$$

Sketch, on separate diagrams, showing clearly any asymptotes, intersections with the axes and coordinates of stationary points, the graphs of

(i) $y=\mathrm{f}(x)$, (ii) $y=\mathrm{g}(x)$.

The function h is defined as

$$\mathrm{h}: x \mapsto \mathrm{f}(x), \quad \text{for } x \in \mathbb{R},\ x > -2.$$

(iii) Find the equation of the tangent to $y = \mathrm{gh}(x)$ at the point where $x = 0$.

(iv) Find the area bounded by the graph of $y = \mathrm{gh}(x)$ with the x-axis.

(v) Solve the inequality $\mathrm{gh}(x) < 2\ln(x+1)$.

(b) Sketch the curve $y = 1 + \dfrac{1-10x}{x(x+1)}$, $x \in \mathbb{R}$, $x \neq -1$, $x \neq 0$, stating clearly equations of any asymptotes, intersections with the axes and coordinates of turning points.

(c) Sketch the graph of $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = \begin{cases} \sin x + \cos x - 1 & \text{for } -\frac{7}{12}\pi \le x < 0, \\ -\sqrt{1 - \dfrac{(x-2)^2}{4}} & \text{for } 0 \le x \le 2. \end{cases}$

State clearly the coordinates of the endpoints.

2: Graphing – Parametric Equations

Skills Required: By using the **PARAMETRIC** mode,

1. Sketch the curve for the given domain [Y=, WINDOW].
2. Determine the coordinates of a point on the curve [TRACE, [CALC] value].
3. Determine the gradient of the curve at a point [[CALC] $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ OR nDeriv].

Exercise 2

A curve has parametric equations $x = \sin 2\theta + \sin\theta$, $y = \cos\theta$, where $0 \le \theta \le \dfrac{2\pi}{3}$.

(i) Find the equation of the tangent to the curve at the point where $\theta = \dfrac{\pi}{2}$.

(ii) Sketch the curve, giving the coordinates of any intercepts.

3: Equations & Inequalities

Skills Required:

1. Solve equations & inequalities.

- By using $\boxed{\text{APPS}}$→**PlySmlt2**→SIMULTANEOUS EQN SOLVER,

2. Solve system of linear equations.

Exercise 3

(a) Find the smallest positive integer n such that $10000\left(1-0.9^n\right)<25n(n+3)$.

(b) Find, for $-2\le x\le 2$, the set of values of x such that $\left|e^x\sin x-\left(x+x^2+\frac{x^3}{3}\right)\right|<0.05$.

(c) Use your calculator to find a vector equation of the line of intersection between
$\pi_1:\ \mathbf{r}\cdot(\mathbf{i}+3\mathbf{j}+2\mathbf{k})=4$ and $\pi_2:\ \mathbf{r}\cdot(\mathbf{i}-\mathbf{j}-\mathbf{k})=4$.

(d) A finite arithmetic progression $u_1, u_2, u_3, \ldots, u_{2n}$ has first term a and common difference d.

It is given that $\frac{u_{n+1}}{u_n}=\frac{4027}{4025}$, $u_n+u_{2n}=36228$ and the average of all the odd-numbered terms is 12 075. By formulating three equations involving a, d and nd, find the values of a, d and n.

4: Complex Numbers

Skills Required:

1. Perform operations on complex numbers.

- By using $\boxed{\text{APPS}}$→**PlySmlt2**→POLYNOMIAL ROOT FINDER,

2. Find the roots of a polynomial equation with real coefficients.

- By using $\boxed{\text{MATH}}$→**CMPLX**,

3. Find the conjugate, modulus and argument of a complex number.

Exercise 4

(a) Evaluate the following: (i) $\frac{(3+4\mathrm{i})^3}{1-\mathrm{i}}$, (ii) $\sqrt{5-12\mathrm{i}}$. [(i) $-80.5-36.5\mathrm{i}$; (ii) $3-2\mathrm{i}$]

(b) Find the roots of the following equations:
(i) $z^2+(-1+4\mathrm{i})z+(-5+\mathrm{i})=0$, (ii) $z^4-2z^3+8z-16=0$.

5: Discrete Random Variables

Skills Required: Use of [STAT], summation, [TABLE].

Exercise 5

(a) Find the mean and standard deviation of the random variable X with the following probability distribution:

x	-4	-3	-2	-1	0	1	2	3
$\text{P}(X = x)$	0.04	0.16	0.24	0.16	0.15	0.1	0.1	0.05

(b) A committee of 12 people is chosen at random from a group consisting of 24 women and 16 men. The number of women on the committee is denoted by R.

Use your calculator to find
(i) $\text{P}(R \geq 6)$,
(ii) the probability that R is divisible by 3,
(iii) $\text{E}(R)$,
(iv) $\text{Var}(R)$,
(v) the most probable number of women on the committee.